CSI ORIGIN 2026

PROBLEM STATEMENT

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Hackathon / Event Name: CSI ORIGIN 2026

Problem Statement ID: **5**

Title: Building a Competitive Capital Market for Supply-Chain Working Capital

2 BACKGROUND

Current Situation / Context

Supply-chain financing is often constrained by fragmented access to capital. A supplier with a legitimate invoice may approach a single bank or financing provider, resulting in limited price discovery, rigid financing terms, and inefficient allocation of available capital.

At the same time, banks, NBFCs, funds, fintechs, and other capital providers may have access to deployable capital but limited visibility into high-quality supply-chain financing opportunities. This creates a structural mismatch between financing demand and available capital.

Traditional supply-chain financing therefore commonly follows a bilateral model in which a supplier approaches an individual financier for a financing arrangement rather than accessing a competitive pool of capital.

Underlying Problem or Need

The core gap is the absence of an intelligent and competitive financing marketplace capable of connecting verified supply-chain financing demand with multiple capital providers.

A supplier should be able to present a verified invoice to a financing market where eligible capital providers can independently evaluate the opportunity and compete to provide financing under terms suited to the supplier's requirements and the provider's risk and capital constraints.

Why This Is Becoming Important

Supply-chain working-capital requirements vary across suppliers, industries, payment cycles, and business conditions. At the same time, capital providers differ in liquidity availability, risk appetite, pricing, financing capacity, and return requirements.

A bilateral financing model limits the ability of these participants to discover mutually advantageous terms. A continuously operating marketplace can improve price discovery and allow capital to be dynamically allocated toward financing opportunities where it can be deployed most effectively.

Limitations of the Current Situation

In traditional financing arrangements, suppliers may have limited visibility into alternative financing offers and may lack meaningful negotiating power over rates, fees, advance rates, tenor, and settlement terms.

Capital providers may simultaneously struggle to identify suitable financing opportunities that match their risk appetite, available liquidity, and portfolio constraints.

Simply presenting multiple loan offers also does not solve the underlying problem, because the most attractive financing option for a supplier may not be the offer with the lowest interest rate. Advance rate, fees, tenor, settlement speed, repayment structure, and supplier requirements may materially affect the overall value of an offer.

Impact of Leaving This Unresolved

If financing remains fragmented and predominantly bilateral, suppliers may continue to face limited financing choices, inefficient pricing, delayed access to working capital, and rigid financing terms.

Capital providers may also fail to efficiently deploy available liquidity, resulting in missed financing opportunities and inefficient allocation of capital across the supply chain.

3 PROBLEM

What Is Going Wrong / Missing?

The current supply-chain financing model lacks a competitive, continuously clearing marketplace capable of intelligently matching verified financing demand with suitable sources of capital.

The missing capability is not simply a platform that displays competing financing offers. The system must assess financing opportunities, understand the requirements and constraints of both suppliers and capital providers, determine which opportunities should be surfaced to which providers, compare competing offers, and identify the financing outcome that provides the best overall fit.

Who or What Is Affected?

The problem affects suppliers seeking working-capital financing against legitimate invoices, as well as banks, NBFCs, funds, fintechs, and other capital providers seeking suitable opportunities to deploy capital.

It also affects the broader supply chain, where the availability, cost, and timing of working-capital financing can influence supplier liquidity and the efficiency of commercial relationships.

Why Is It Difficult or Important?

The marketplace must operate under information asymmetry, incomplete information, changing liquidity, varying risk appetites, and competing objectives.

For suppliers, the optimal financing offer is not necessarily the one with the lowest financing rate. A higher-cost offer may provide a greater advance rate, faster settlement, better tenor, or lower fees and therefore provide greater overall value.

For capital providers, financing decisions must account for risk-adjusted returns, available liquidity, portfolio constraints, and the characteristics of the supplier and buyer.

The system must therefore solve a dynamic matching and allocation problem rather than a simple offer-comparison problem.

Consequences

If this problem persists, suppliers may continue to depend on limited financing relationships and receive suboptimal financing terms or delayed access to working capital.

Capital providers may have difficulty discovering suitable opportunities and deploying capital efficiently, while the broader supply chain may experience avoidable liquidity constraints and inefficient financing allocation.

4 CHALLENGE

Design and build a **competitive, agentic supply-chain financing marketplace** that enables verified invoices to be matched with multiple competing capital providers and determines financing outcomes based on the requirements, risk constraints, liquidity, and objectives of both sides.

The system should continuously analyse financing demand and available capital, intelligently surface relevant opportunities to capital providers, generate or evaluate competing financing offers, and determine the most suitable financing outcome for the supplier.

The system should operate as a continuous allocation loop:

Invoice → Verify → Assess Risk → Discover Capital → Generate Offers → Compare → Match → Finance → Settle → Learn

Problem Requirements / Constraints

1. The system must enable verified supply-chain invoices to be presented as financing opportunities to multiple eligible capital providers.
2. The system must intelligently match financing opportunities with capital providers based on relevant factors such as risk appetite, available liquidity, financing capacity, and supplier or buyer characteristics.
3. The system must support competing financing offers that can differ across interest or financing rate, tenor, advance rate, fees, settlement speed, repayment structure, and other relevant terms.
4. The system must evaluate financing offers based on overall suitability rather than relying solely on the lowest financing rate.
5. The system must account for information asymmetry, incomplete information, changing capital availability, and differing risk profiles when determining financing matches.
6. The system must enable capital providers to evaluate opportunities against risk-adjusted return requirements and portfolio or capital constraints.
7. The system must support a complete financing workflow covering invoice verification, risk assessment, capital discovery, offer generation, matching, financing, settlement, and subsequent learning or reallocation.

5 ANNEXURE — DETAILED RULES / CONSTRAINTS

Agent Autonomy

The system should minimise the need for continuous human intervention across task discovery, participation, verification, and settlement. Human involvement may exist where necessary for exceptional cases or dispute escalation.

Invoice Verification

Financing opportunities must be based on invoices or financing claims that have undergone an appropriate verification process. The system should distinguish verified information from uncertain or incomplete information when evaluating opportunities.

Capital Provider Matching

The system must account for differences among capital providers, including available liquidity, risk appetite, financing capacity, return requirements, and portfolio constraints. A financing opportunity should be surfaced to providers for whom the opportunity is reasonably suitable.

Offer Evaluation

Financing offers must be evaluated across multiple dimensions rather than comparing interest rates alone. Relevant considerations may include financing rate, tenor, advance rate, fees, settlement speed, repayment structure, and the supplier's specific liquidity requirements.

Risk and Information Asymmetry

The system must account for incomplete information and uncertainty regarding supplier, buyer, invoice, and repayment characteristics. Risk assessment should influence both financing eligibility and the suitability of capital-provider matches.

Supplier Outcomes

The system should optimise for the supplier's overall financing outcome rather than a single pricing metric. Financing cost, amount of capital received, settlement speed, tenor, fees, and other relevant terms should be considered together.

Capital Provider Objectives

Capital allocation decisions must account for the objectives and constraints of capital providers, including risk-adjusted returns, available liquidity, portfolio concentration, and financing capacity.

Dynamic Marketplace

The marketplace should be capable of responding to changes in financing demand, available capital, risk conditions, and financing terms. Matching and allocation should therefore be treated as dynamic rather than as a one-time static process.

Settlement Reliability

The system must account for the reliability of financing and settlement processes. A financing match should not be considered complete merely because an offer has been selected; the relevant financing and settlement conditions must be satisfied.

6 FINAL PROBLEM CONSTRAINTS

— Financing opportunities must be based on appropriately verified supply-chain invoices or financing claims.
— The system must support multiple capital providers competing to finance suitable opportunities.
— Matching must account for both supplier requirements and capital-provider constraints, including risk appetite, liquidity, financing capacity, and portfolio considerations.
— Financing offers must be evaluated across multiple dimensions rather than solely by interest or financing rate.
— The system must account for information asymmetry, incomplete information, changing liquidity, and risk-based pricing.
— The system must optimise for overall financing suitability for the supplier while maintaining risk-adjusted capital allocation for providers.
— The system must support a dynamic financing lifecycle from verification through matching, financing, settlement, and subsequent reallocation or learning.
— The challenge is focused on creating an intelligent competitive capital market for supply-chain working capital, not merely a loan-comparison platform or static financing marketplace.